

Sprint Report – Sprint 5

Sprint Goal

Strengthen the production readiness of the ML system by completing CI/CD setup, improving testing, finalizing documentation, and validating end-to-end pipeline execution.

Sprint Duration

30 March – 5 April 2026

Team Roles

- Faezeh & Sepideh & Adi: CI/CD, testing, documentation, pipeline validation
 - Adi: Scrum master
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Sprint Backlog & Progress

Completed Tasks

- [x] Task 20 — Technical Design Documentation
 - [x] Task 19 — Analysis Documentation
 - [x] Task 22 — Implement Data Versioning (final refinements)
 - [x] Task 23 — Orchestrate ML Pipeline Using Airflow (stabilization)
 - [x] Task 24 — Dockerize ML Inference Service (improvements)
 - [x] CI/CD Skeleton
 - [x] E2E Slice 1 for Pipeline
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In Progress / To Do

- [] Unit Tests in CI
 - [] Documentation for CI/CD
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Key Deliverables

1. Technical & Analysis Documentation

- Completed Analysis Document
 - Data understanding
 - Feature engineering decisions
 - Modeling challenges and limitations
 - Completed Technical Design Document
 - System architecture
 - Tooling justification (Airflow, MLflow, FastAPI, DVC)
 - Pipeline structure and deployment setup
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2. CI/CD Pipeline (Initial Setup)

- Created CI/CD skeleton
- Defined structure for:
- Automated builds
- Pipeline execution checks
- Future test integration

➡■ Foundation for full automation is now in place

3. End-to-End Pipeline Validation

- Implemented E2E Slice 1
 - ingestion → feature engineering → modeling → prediction
 - Ensures:
 - Full system integration
 - Pipeline runs end-to-end
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4. System Stabilization

- Improved:
 - Airflow orchestration
 - Docker inference service
 - Data versioning consistency
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Key Insights

- System integration validated end-to-end
- CI/CD foundation ready but incomplete
- Testing is the main gap

➡■ Next step: automated testing + CI/CD completion

Sprint Metrics

Cycle Time (Estimated)

Task Group	Duration
Documentation	2–3 days
CI/CD Setup	1–2 days
E2E Validation	1–2 days
System Stabilization	1–2 days

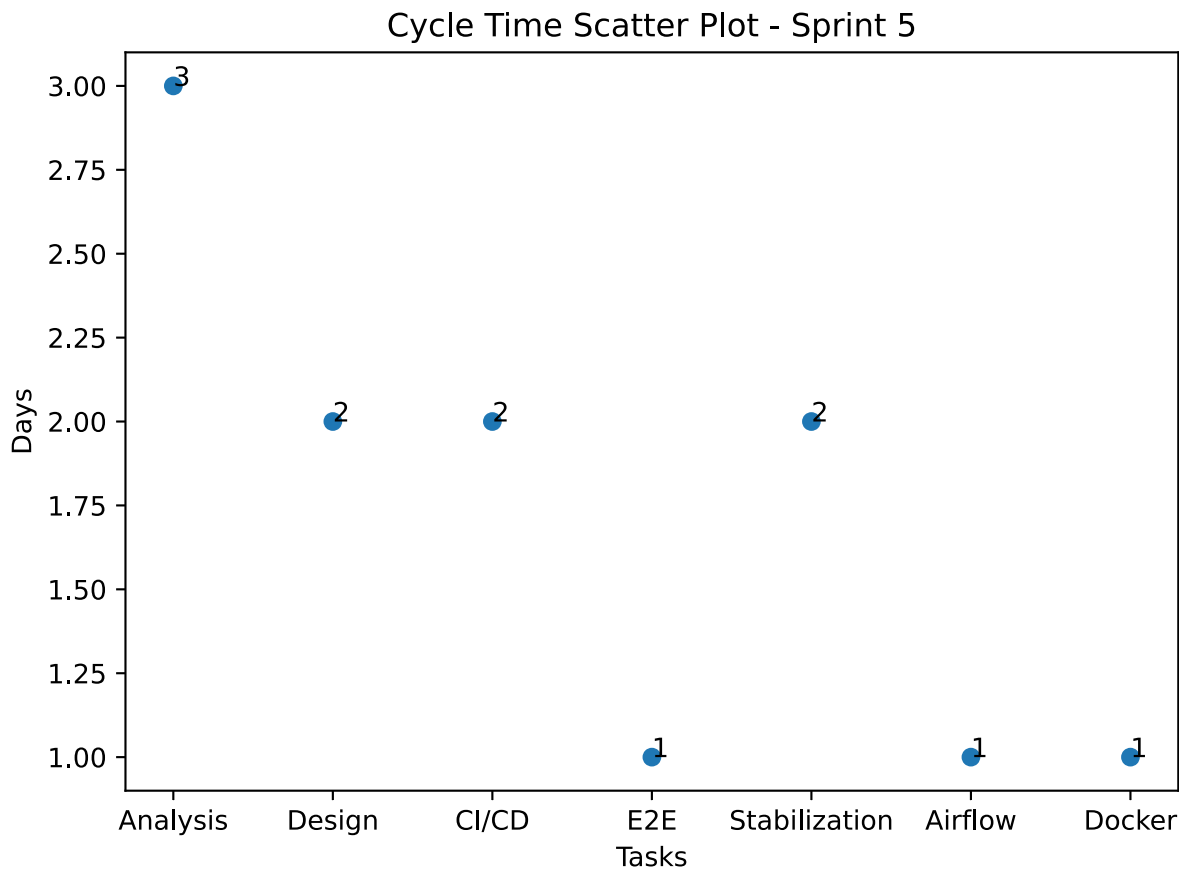
Average Cycle Time: ~2 days

Total Tasks Completed: 7

Completion Rate: ~80–85%

Cycle Time Analysis

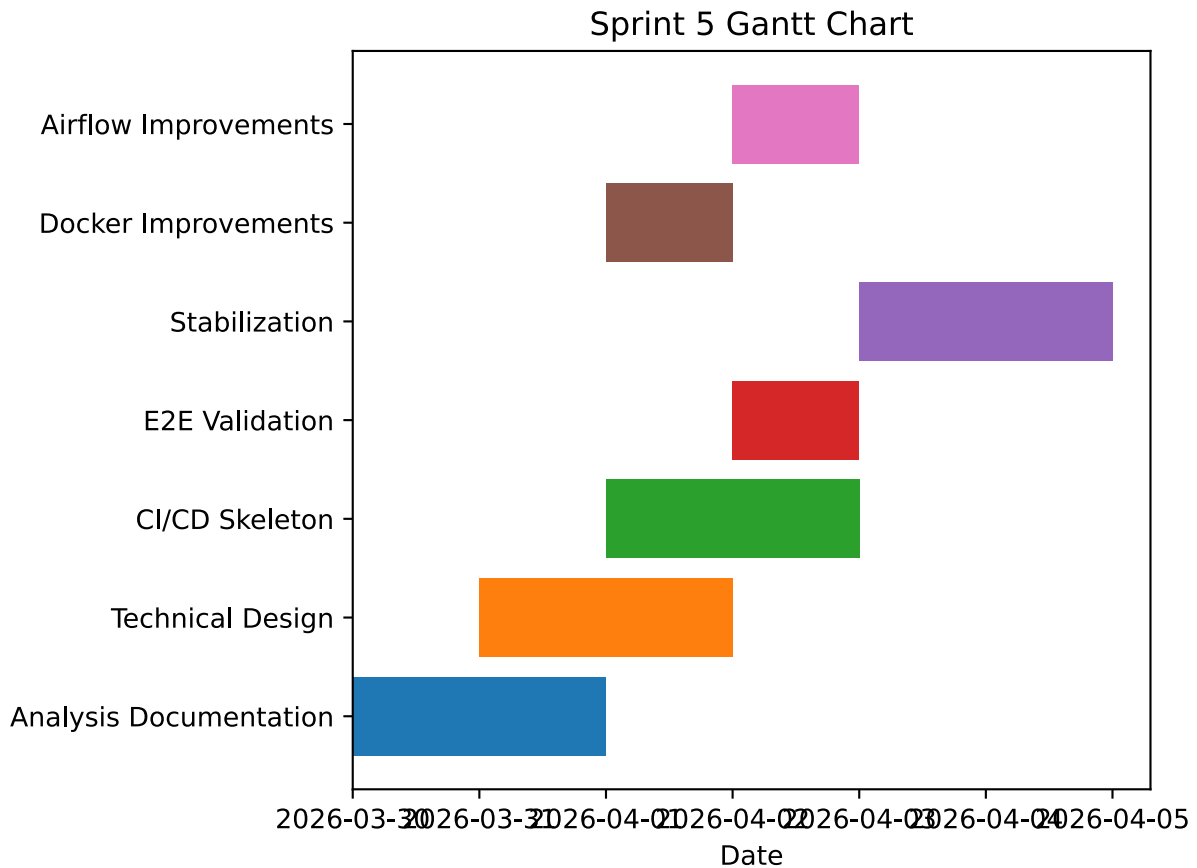
Cycle Time Scatter Plot



Cycle time scatter plot

This chart shows that most tasks were completed within 1–3 days, indicating stable development and manageable complexity.

Cycle Time Gantt Chart



Cycle time Gantt

The Gantt chart shows the progression of work during the sprint:

- Early sprint → documentation
- Mid sprint → CI/CD and integration
- Late sprint → stabilization and testing

Sprint Summary

Sprint 5 focused on closing the gap between development and production readiness.

Key achievements:

- Completed analysis & design documentation
- Established CI/CD foundation
- Validated end-to-end pipeline execution

Main challenge:

- CI/CD not fully complete yet (testing pending)